



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

SPECTROSCOPY OF ASTROPHYSICAL PLASMAS

Fritz Ali Agildere

June 9, 2026



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.

Overview

Physical Properties

- Galaxy Clusters
- Radiation Mechanisms

Radial Profiles

- Data Processing
- Theory & Results

Observations

- Detectors & Telescopes
- Spectral Characteristics
- Conclusion & Outlook

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Cluster Models & Classes

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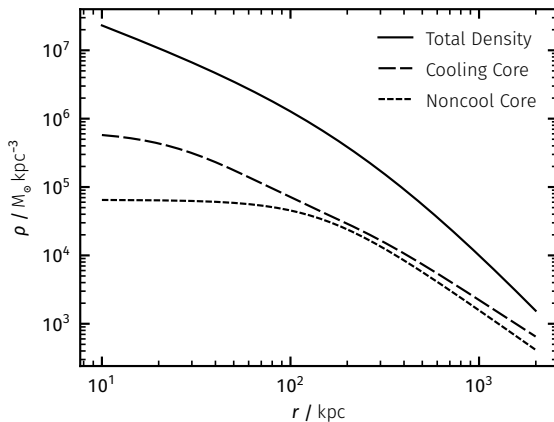
- Sparse & Hot Plasma Systems

Cluster Models & Classes

- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)

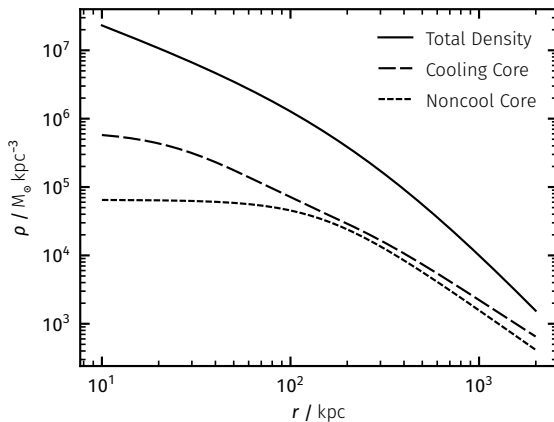
Cluster Models & Classes

- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)
- Purely Gravitational Baseline Assumption



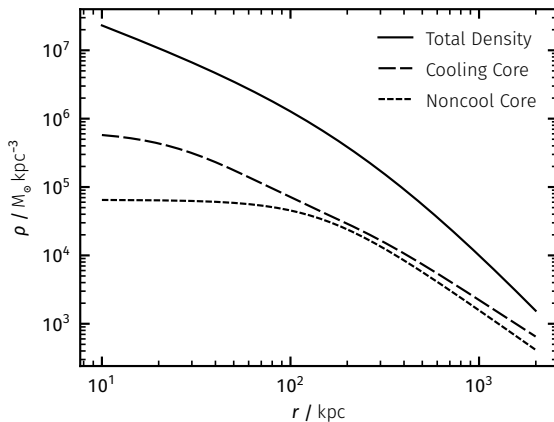
Cluster Models & Classes

- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)
- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Objects



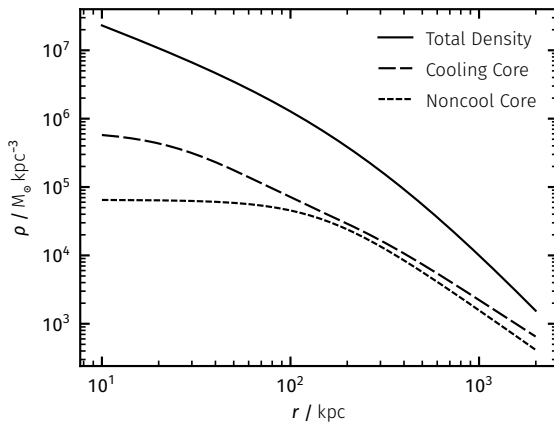
Cluster Models & Classes

- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)
- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Objects
- Selfsimilar Radial Scaling



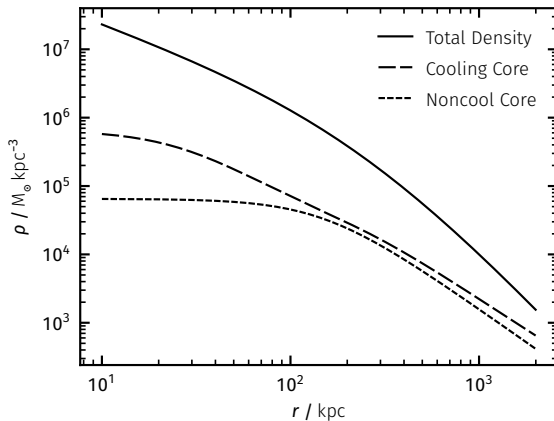
Cluster Models & Classes

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- Selfsimilar Radial Scaling
- Deviation Defines Categories



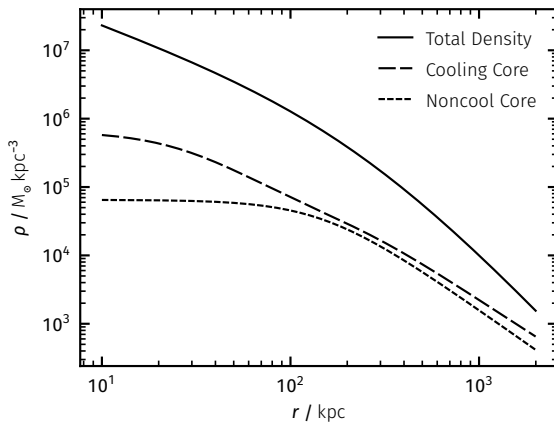
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 - Cooling Cores (CC)



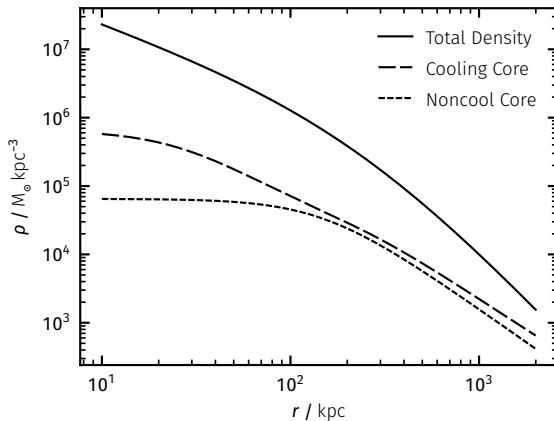
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 - Cooling Cores (CC)
 - Noncool Cores (NCC)



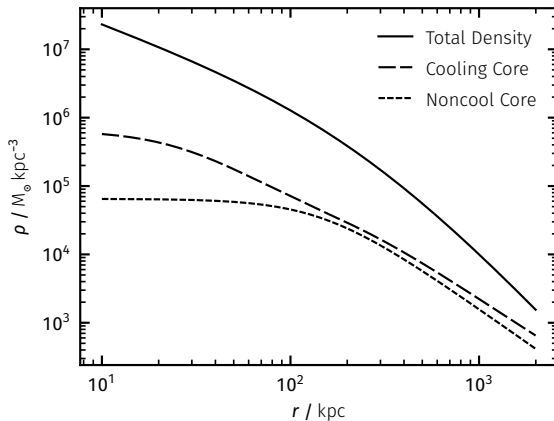
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- Mergers Inject Energetic Shocks & Sloshing
- Central Active Galactic Nuclei (AGN) Feedback



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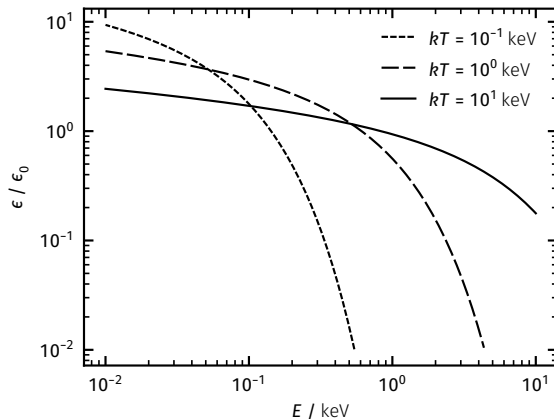
Characteristic Emission

Characteristic Emission

- Bremsstrahlung (*free-free*)

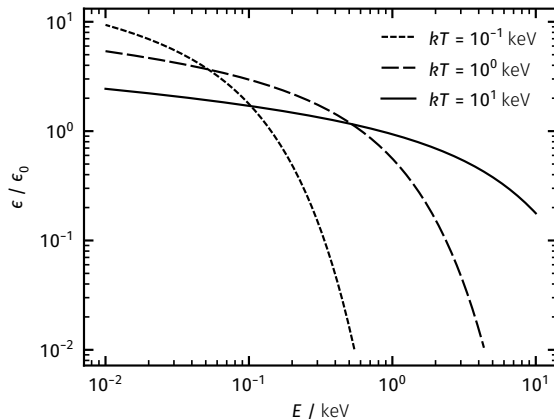
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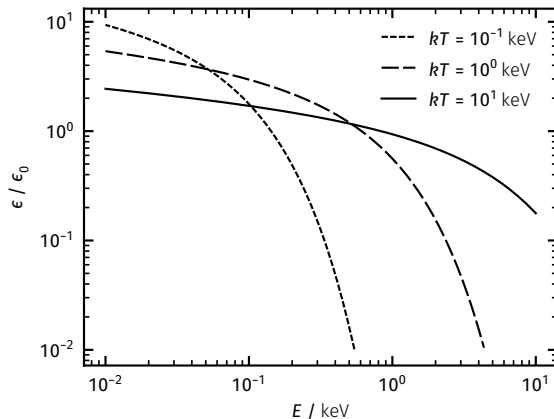
Characteristic Emission

- Bremsstrahlung (*free-free*)
- Broad Continuous Spectrum



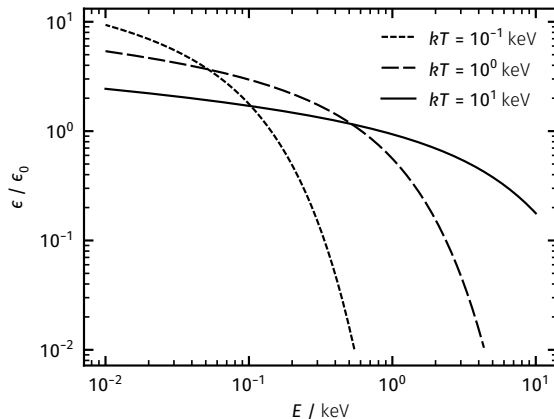
Characteristic Emission

- Bremsstrahlung (*free-free*)
 - Broad Continuous Spectrum
- Recombination (*bound-free*)



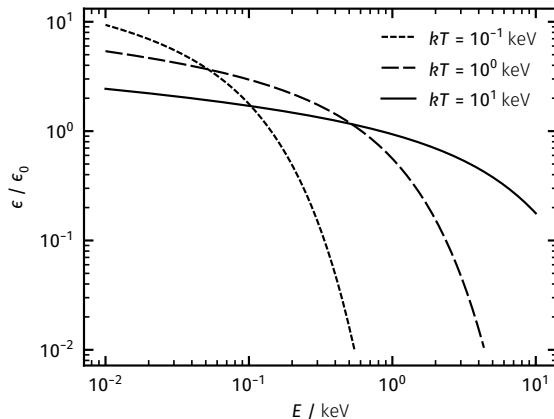
Characteristic Emission

- Bremsstrahlung (*free-free*)
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- Recombination (*bound-free*)
 - Soft Continuum



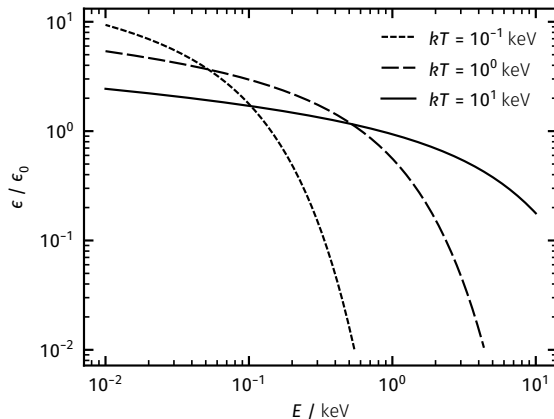
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- Recombination (*bound-free*)
 - Soft Continuum
 - Hard Ionization Edge



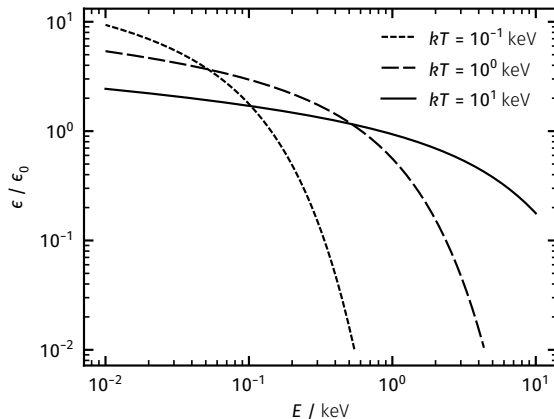
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 - Hard Ionization Edge
- Atomic Transitions (*bound-bound*)



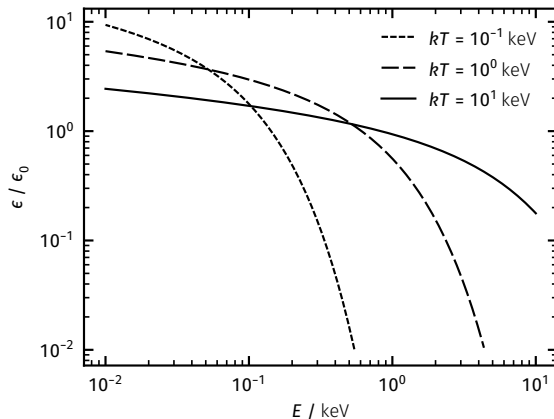
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 - Discrete Lines



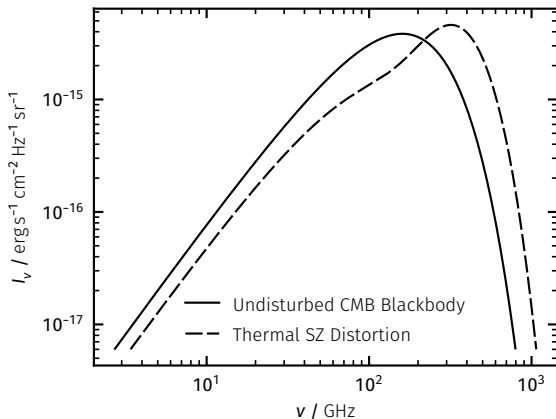
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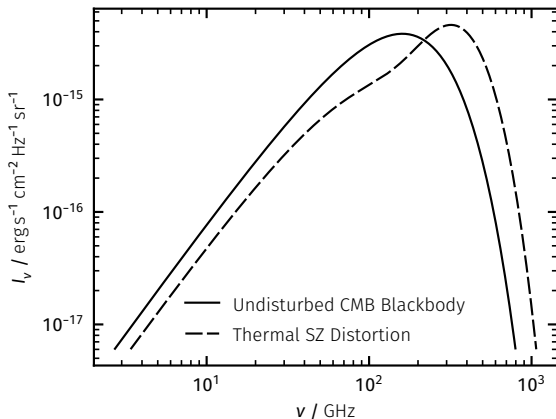
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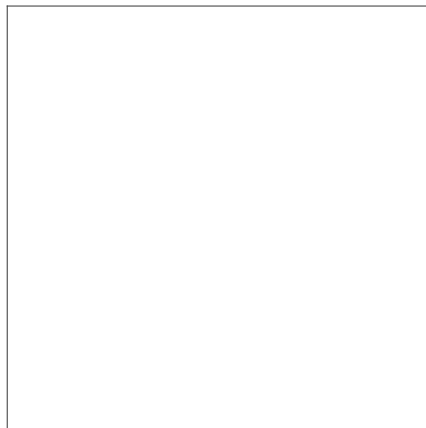
Extracting Information

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- Annulus Decomposition

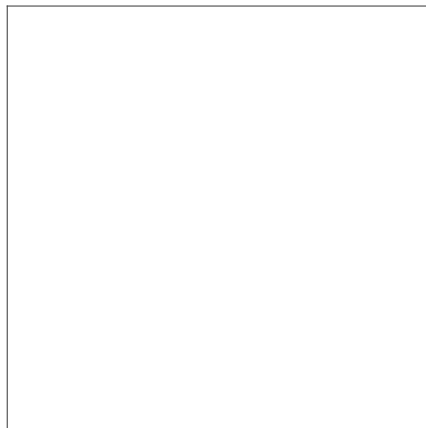
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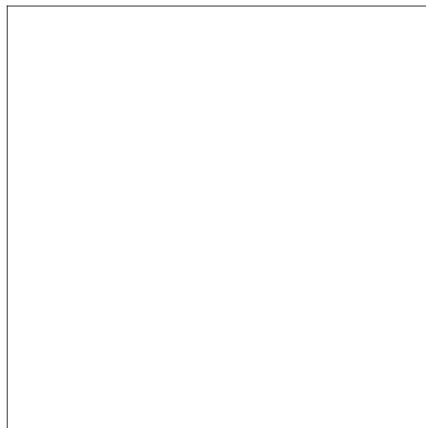
Extracting Information

- Annulus Decomposition
- Projection Unfolding



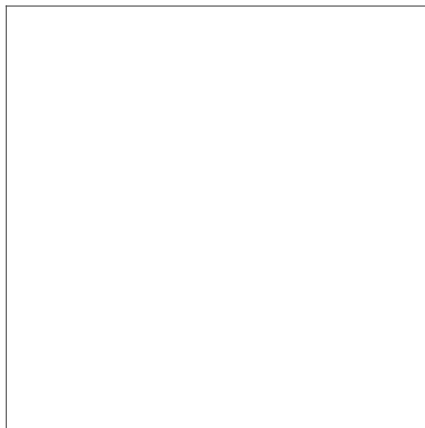
Extracting Information

- Annulus Decomposition
- Projection Unfolding
- Spectral Fitting



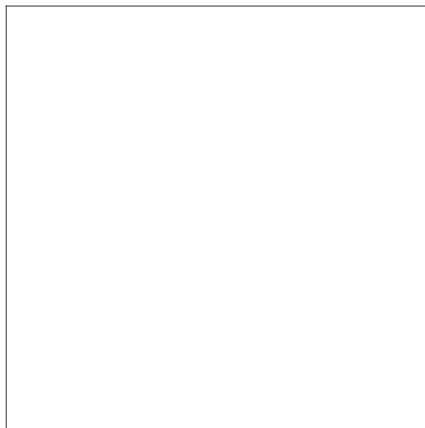
Extracting Information

- Annulus Decomposition
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 - Obtain Temperature, Density, Abundances, Redshift



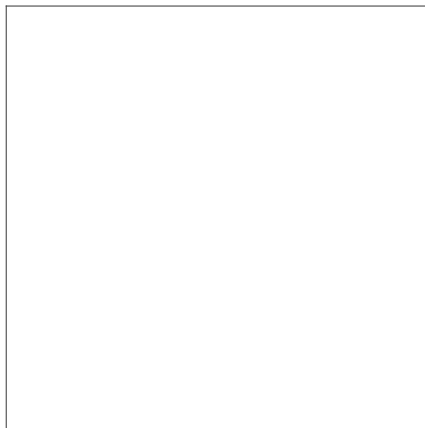
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 - Obtain Temperature, Density, Abundances, Redshift
 - Derive Pressure, Entropy, Mass



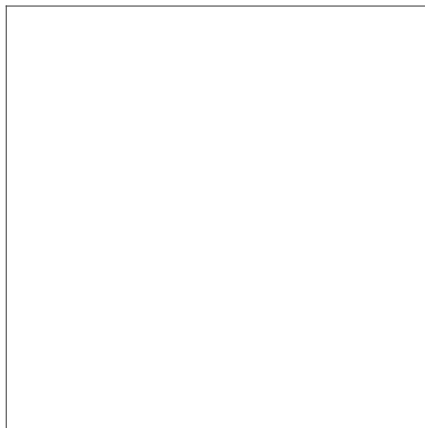
Extracting Information

- Annulus Decomposition
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 - Obtain Temperature, Density, Abundances, Redshift
 - Derive Pressure, Entropy, Mass
- Spherically Symmetric Averaging



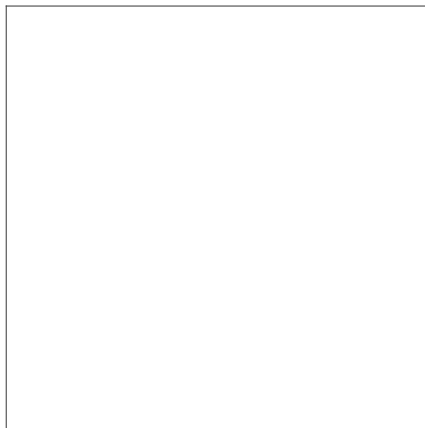
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- Spherically Symmetric Averaging
- Disturbed Systems Breakdown Possible



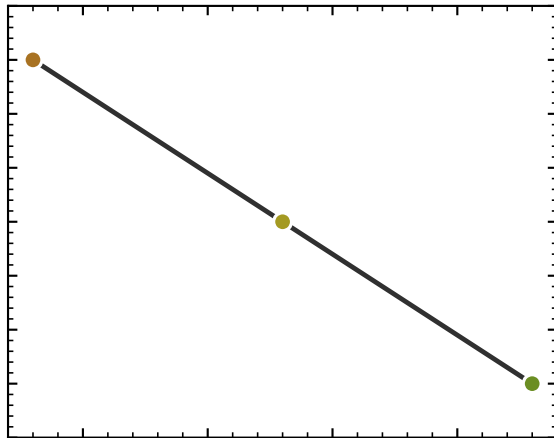
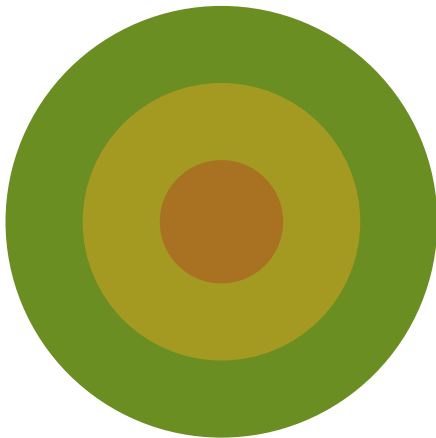
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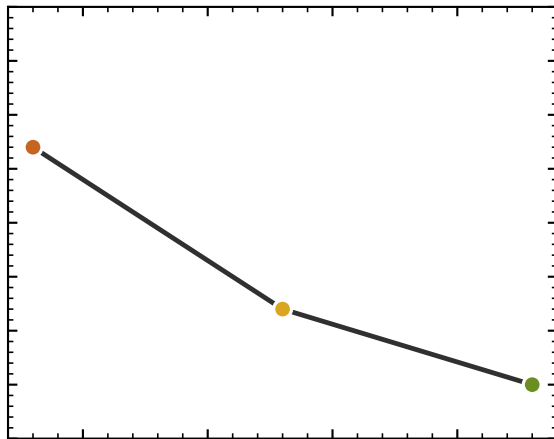


Deprojection

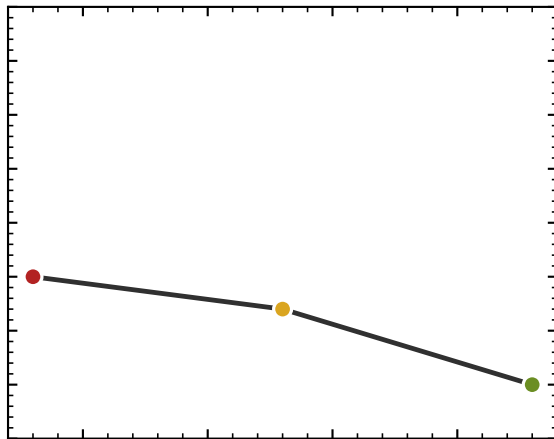
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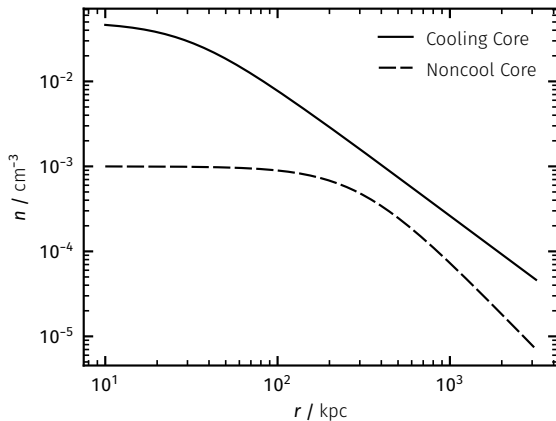
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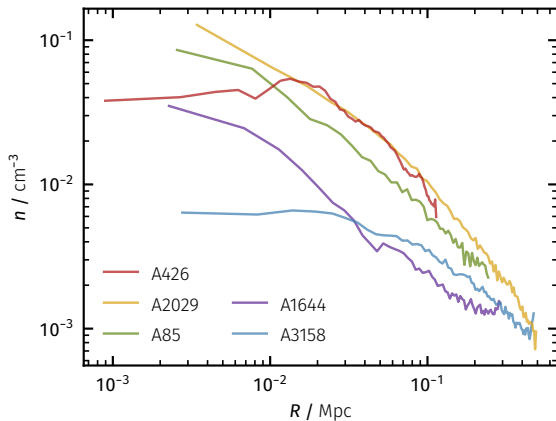
Conclusion & Outlook

Density & Pressure

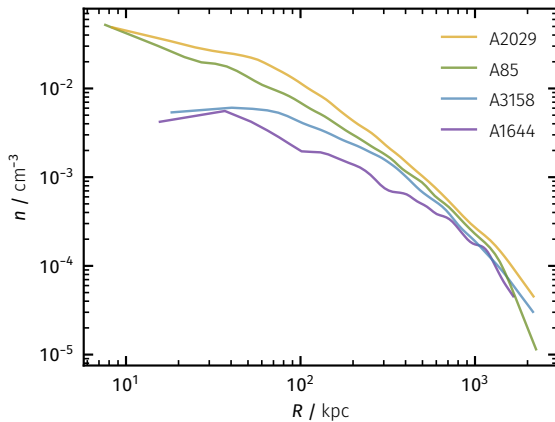
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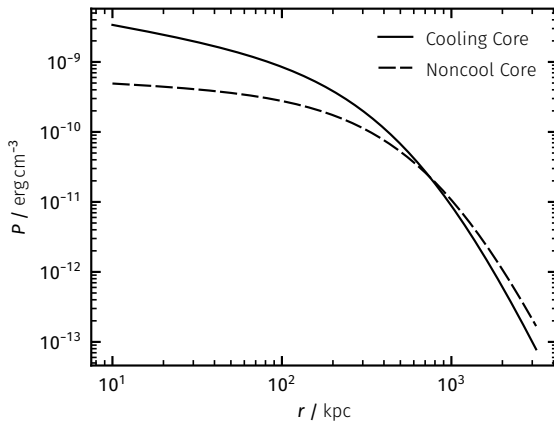
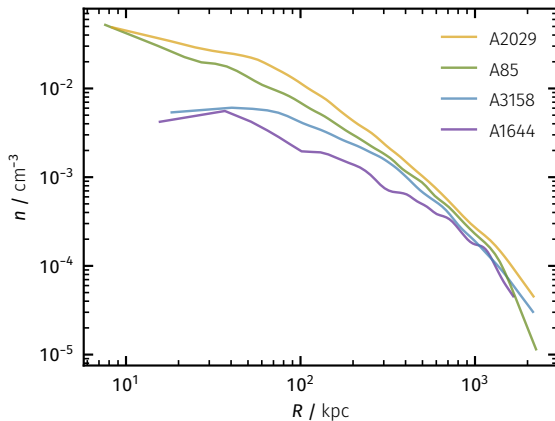
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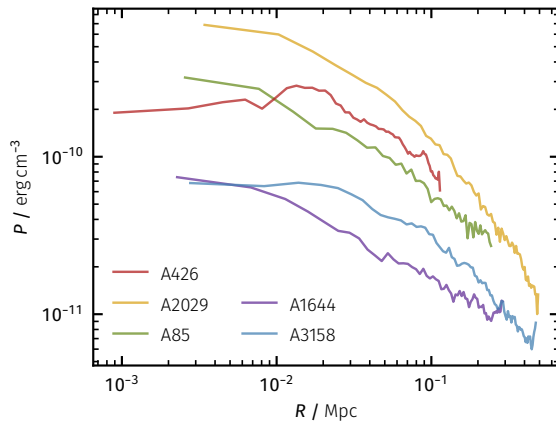
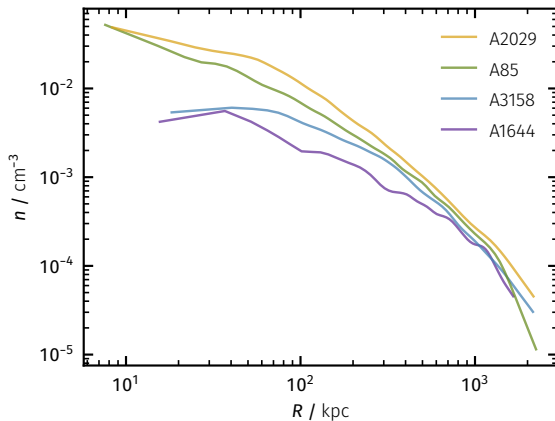
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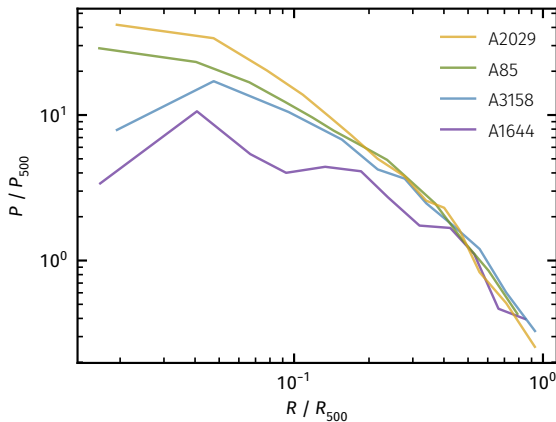
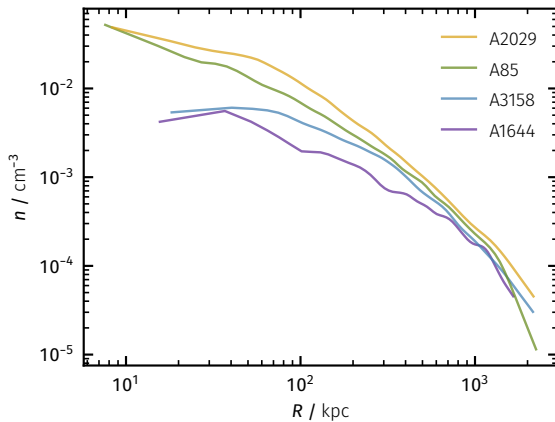
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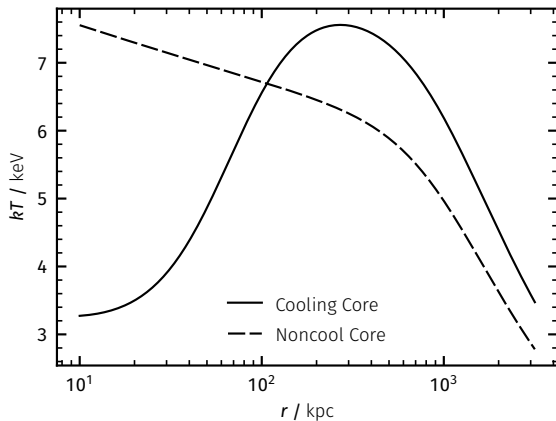


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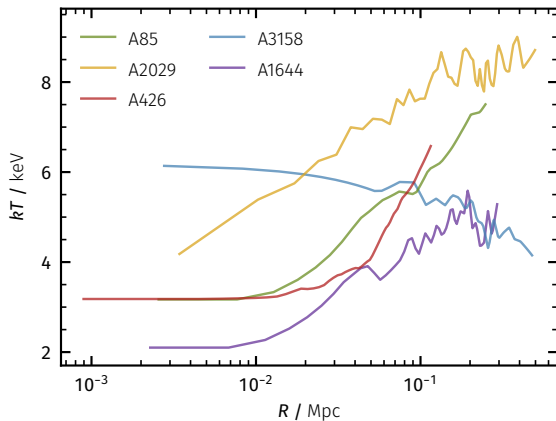


Temperature & Entropy

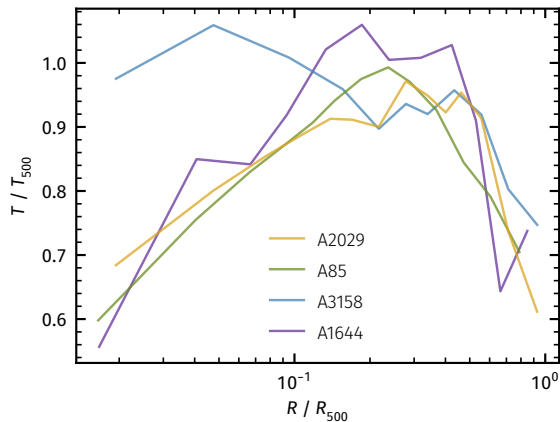
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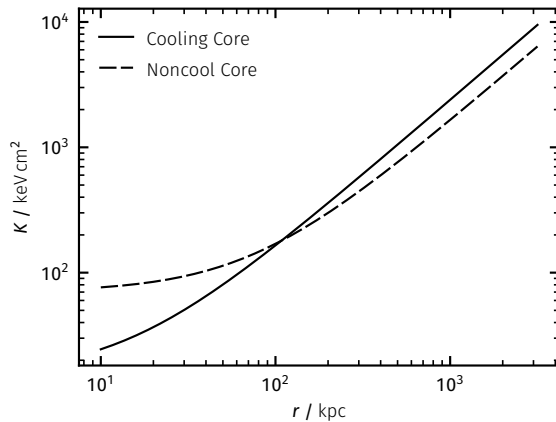
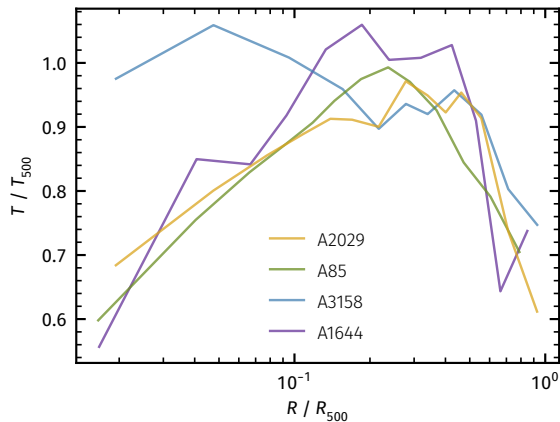
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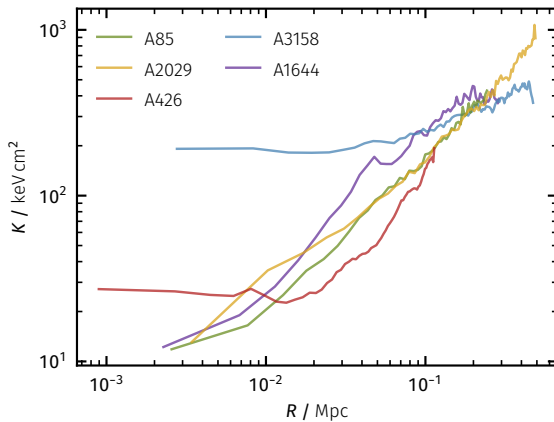
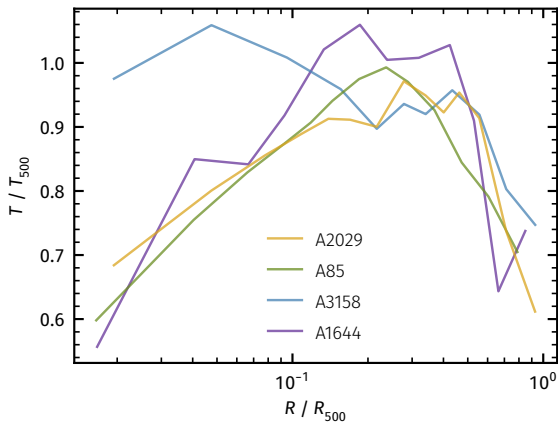
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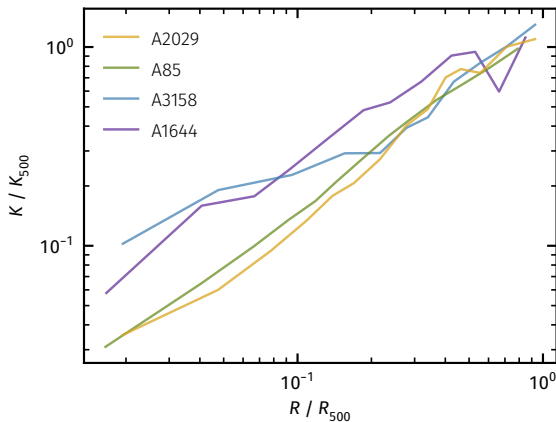
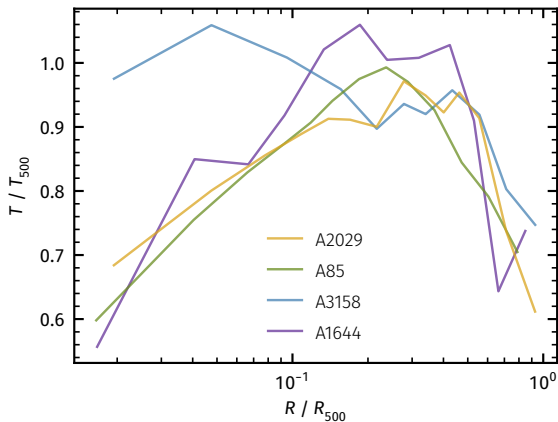
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Selected Missions

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- Large Survey

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 - ROSAT (*retired*)

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- Large Survey
 - ROSAT (*retired*)
 - Planck (*retired*)
 - eROSITA (*active*)



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- Large Survey
 - ROSAT (*retired*)
 - Planck (*retired*)
 - eROSITA (*active*)
- Spatial Resolution



Selected Missions

- Large Survey
 - ROSAT (*retired*)
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 - eROSITA (*active*)
- Spatial Resolution
 - Chandra (*active*)



Selected Missions

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 - ROSAT (*retired*)
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 - eROSITA (*active*)
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 - Chandra (*active*)
 - NewAthena (*planned*)



Selected Missions

- Large Survey
 - ROSAT (*retired*)
 - Planck (*retired*)
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- Spatial Resolution
 - Chandra (*active*)
 - NewAthena (*planned*)
- Spectroscopic Precision



Selected Missions

- Large Survey
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 - Planck (*retired*)
 - eROSITA (*active*)
- Spatial Resolution
 - Chandra (*active*)
 - NewAthena (*planned*)
- Spectroscopic Precision
 - ASCA (*retired*)



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 - eROSITA (*active*)
- Spatial Resolution
 - Chandra (*active*)
 - NewAthena (*planned*)
- Spectroscopic Precision
 - ASCA (*retired*)
 - XMM-Newton (*active*)



Selected Missions

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 - ROSAT (*retired*)
 - Planck (*retired*)
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 - XRISM (*active*)



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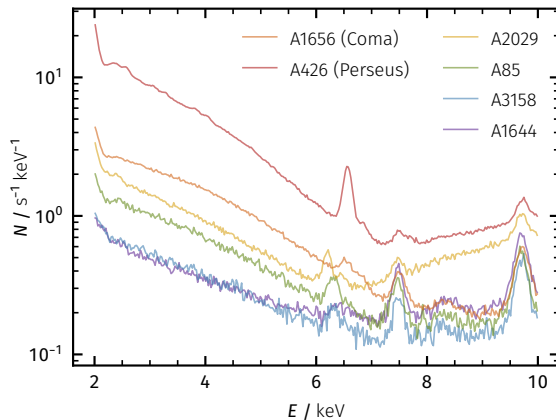
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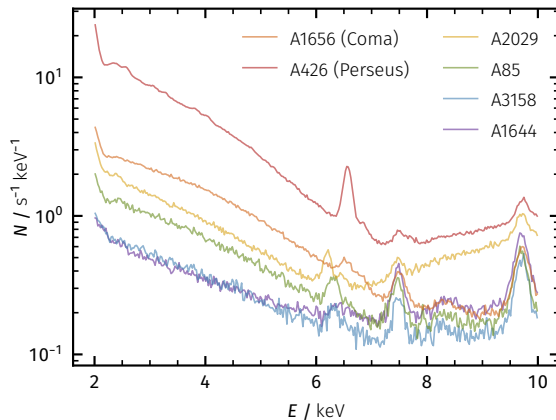
Spectrum Analysis

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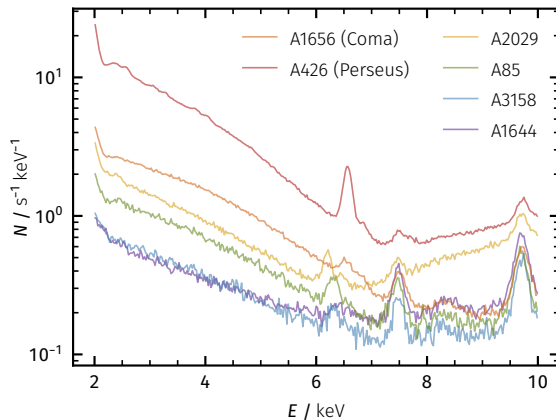
Spectrum Analysis

■ Continuum



Spectrum Analysis

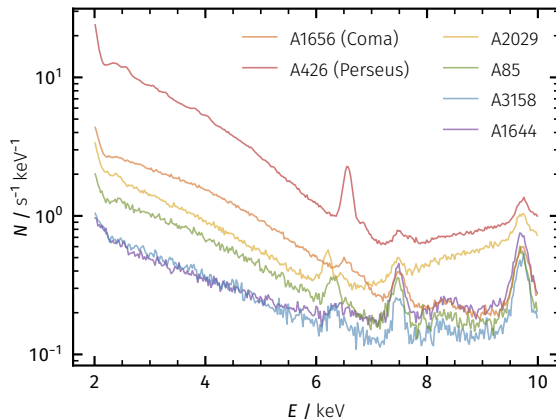
- Continuum
- Bremsstrahlung Baseline



Spectrum Analysis

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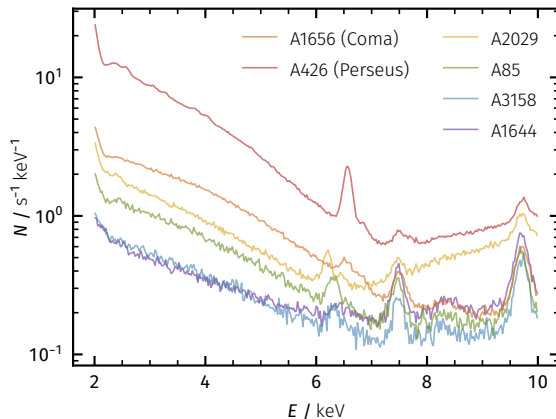
- Bremsstrahlung Baseline
- Possible Soft & Hard Excesses



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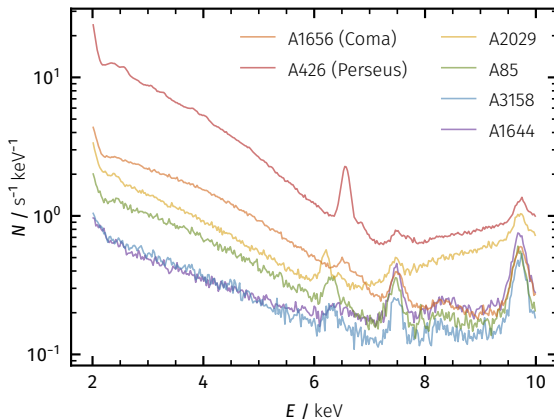
- Bremsstrahlung Baseline
- Possible Soft & Hard Excesses
 - Multiphase Intracluster Gas



Spectrum Analysis

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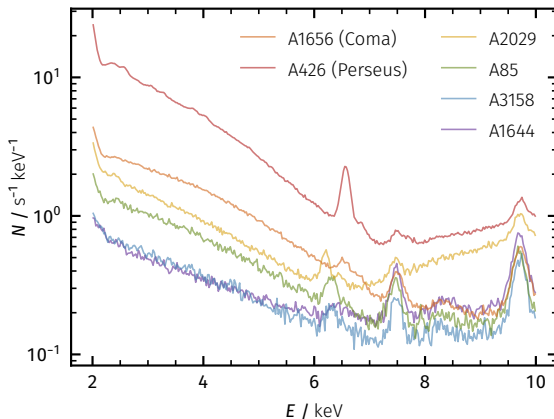
- Bremsstrahlung Baseline
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 - Multiphase Intracluster Gas
 - Warm-Hot Intergalactic Medium



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 - Background Upscattering

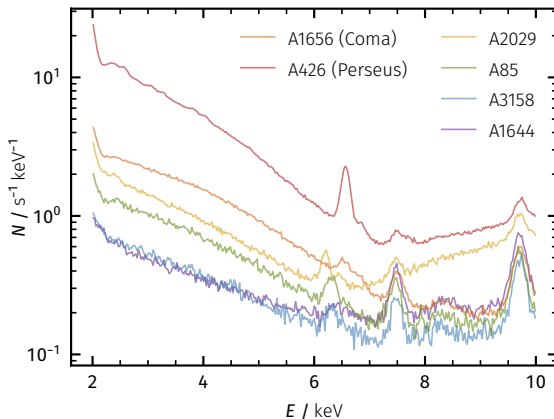


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■ Line Emission



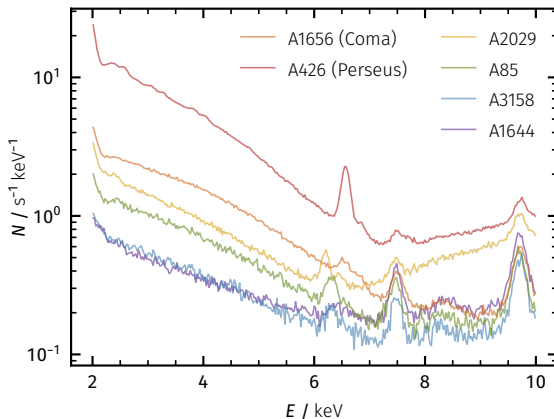
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■ Line Emission

- Powerful Flux Ratio Analyses



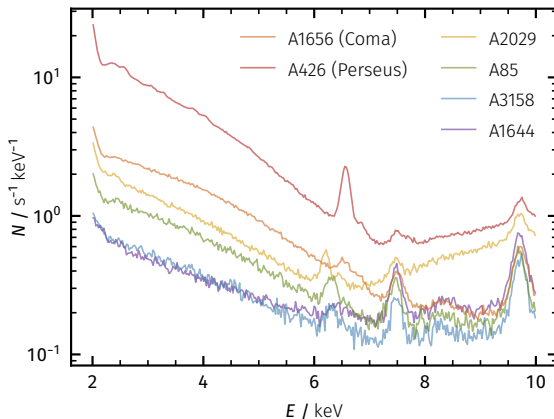
Spectrum Analysis

■ Continuum

- Bremsstrahlung Baseline
- Possible Soft & Hard Excesses
 - Multiphase Intracluster Gas
 - Warm-Hot Intergalactic Medium
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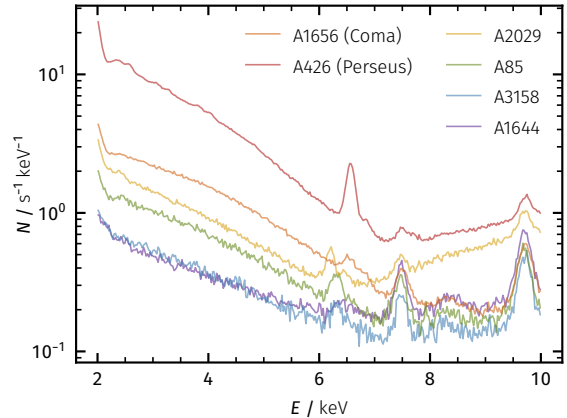
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Overview

Physical Properties

Galaxy Clusters

Radiation Mechanisms

Radial Profiles

Data Processing

Theory & Results

Observations

Detectors & Telescopes

Spectral Characteristics

Conclusion & Outlook

Discussion

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- New Simulation & Data Turbulence Comparison

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)

Appendix